

# CICIL: Empowering Indonesian students with innovative financing built on cloud solutions

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## ABOUT CICIL



## About CICIL

CICIL doubled app deployment speed, driven credit model education events cities remotely, and developed an Asian expansion strategy. Google Cloud and Workspace solutions.

CICIL is an Indonesian fintech startup that empowers young Indonesians with attainable student loans for higher education. The platform deploys AI-driven credit scoring based on nontraditional data, enabling students to win financing without a credit history or bank account. CICIL is licensed under Indonesia's Financial Services Authority OJK and a member of Indonesia's Fintech Association. In 2018, CICIL secured Series A funding led by East Ventures and Vertex Ventures and was selected

## Google Cloud results

- Frees development team to create solutions by slashing DevOps tasks by 50%
- Enhances risk modeling speed and accuracy by enabling AI-driven credit scoring
- Enables full-throttle operations during Covid 19 with zero downtime through productivity and collaboration tools
- Promotes student networks by enabling 25% increase in event

## Doubling of app deployment with migration

as a part of Google Launchpad Accelerator Programme LPA4.

Industries: Financial Services & Insurance

Location: Indonesia

(<https://cloud.google.com/functions/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/monitoring/>), Cloud

BigQuery (<https://cloud.google.com/bigquery/>)

Storage (<https://cloud.google.com/storage/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

Monitoring (<https://cloud.google.com/monitoring/>)

Kubernetes Engine

Cloud SQL (<https://cloud.google.com/sql/>) line/,

TensorFlow (<https://www.tensorflow.org/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Vision API (<https://cloud.google.com/vision/>)

, Google Drive

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

, Google Meet

TensorFlow (<https://www.tensorflow.org/>) /meet/

Google Workspace

(<https://workspace.google.com/>)

Calendar

(<https://workspace.google.com/products/calendar/>)

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participation  
with video  
communication

Google Drive

(<https://workspace.google.com/products/drive/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

According to the World Bank

(<https://ieg.worldbankgroup.org/blog/investing-higher-education-better-development>)

, "higher education is central to creating a knowledge economy and it is critical for economic development in general." Yet in Indonesia, dreams of emerging as a new economic dynamo are being held back, as university remains out of reach for the vast majority in the huge nation of 270 million people.

Only about 16% of Indonesians

([https://www.oecd.org/education/education-at-a-glance/EAG2019\\_CN\\_IDN.pdf](https://www.oecd.org/education/education-at-a-glance/EAG2019_CN_IDN.pdf))

between 25 and 64 have attained higher education, far below the OECD (<https://www.oecd.org/>) average of 44%, and the G20 (<https://www.g20.org/>) average of 38%, underscoring the need for more skilled workers to achieve Indonesia's vast economic potential.

CICIL (<https://www.cicil.co.id/>) is a startup launched by

three friends on a mission to enable Indonesia's aspirational journey with a fintech solution that extends affordable loans to students with neither credit history nor bank account. CICIL deploys AI-driven credit scoring based on unconventional data, such as class attendance and mobile device payments, to accurately model risk for young people in a country where 66% of the population

(<https://www.thinkwithgoogle.com/intl/en-apac/future-of-marketing/machine-learning/all-eyes-e-money-race-reach-180m-unbanked-indonesians/#:~:text=Indonesia%20is%20one%20of%20the,t%20own%20a%20bank%20account>.)  
is unbanked.

CICIL's empowerment story only begins with loans. The startup supports the whole educational journey with seminars, lectures, and one-to-one coaching via a network of 2,500 external student "ambassadors" who mentor peers across the nation.

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Google  
Workspace,  
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reach even in  
Indonesia's most  
remote corners."

—Ricky Jeremiah, CTO,  
CICIL

In order to lift its mission of empowerment to the scale of Indonesia's gigantic archipelago, CICIL sought modernized Kubernetes-based architecture, along with world-class data crunching and AI tools, to bring the gift of higher education to as many young people as possible. As the pandemic hit Indonesia, it became crucial to switch to an entirely remote model for its educational events and transition to a 100% work-from-home environment for staff dispersed across the country.

It found the ideal solutions in a combination of Google Kubernetes Engine (/kubernetes-engine) (GKE), BigQuery (/bigquery), and Google Workspace (<https://workspace.google.com/>), which enabled CICIL's education mission to grow in impact, scale, and creative collaboration even in challenging pandemic circumstances.

"Our mission of extending financing and mentoring springs from our own experience getting a helping hand as students, and we want to give something back," says Ricky Jeremiah, CTO, CICIL. "In these tough times, it's even more important to open higher education opportunities for young Indonesians. Through the unlimited scaling enabled by Google Cloud (<https://cloud.google.com/>) and remote teaching abilities we gain from Google Workspace, we're able to maximize our reach even in Indonesia's most remote corners."

"Everything has become a lot faster due to the power of GKE. And with Cloud Monitoring we immediately know when some kind of issue is happening with one of our instances. The end result is we're able to

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—Ricky Jeremiah, CTO,  
CICIL

## **Opening a world of cloud opportunity with launch of Jakarta Cloud Region**

Indonesian regulations require firms in sensitive sectors such as finance only to use cloud providers that have a data center presence in Indonesia. This limitation meant that CICIL could only rely on on-premises infrastructure. But when Google Cloud launched the [Jakarta Cloud Region](#)

(</blog/products/infrastructure/new-google-cloud-region-in-jakarta-now-open>)

in June 2020, CICIL jumped at the opportunity to move its infrastructure to cloud. It became one of the first Indonesian financial players to apply for (and receive) authorization for cloud migration.

CICIL met the Google Cloud team in Jakarta in June. To its surprise, the cloud adoption for which they waited four years took only a month-and-a-half to execute. By September, the CICIL platform was

hosted 100% in a container-based environment, with GKE as the core solution, supported by Cloud SQL (/sql), Cloud Storage (/storage), Cloud Functions (/functions), and Pub/Sub (/pubsub).

Meanwhile, CICIL deployed a combination of BigQuery and TensorFlow (<https://www.tensorflow.org/>) to build AI-driven credit models with the reams of unconventional student data it was now empowered to crunch.

The seamless cloud migration transformed CICIL's capabilities, enabling rapid app development based on AI insight, as well as unlimited scaling. By transitioning to Kubernetes-based microservice deployment, Jeremiah's team was able to slash DevOps tasks by 50% and thereby double its speed of application launches, meaning faster loan approvals for students.

Jeremiah says one of his favorite solutions is Cloud Monitoring (/monitoring), which enables real-time error reporting in app development. It means engineers can roll back to any stage in the coding process and fix issues before they become hidden kinks that cause post-deployment breakdowns.

"Everything has become a lot faster due to the power of GKE. And with Cloud Monitoring, we immediately know when some kind of issue is happening with one of our instances," says Jeremiah. "The end result is we're able to serve our customers a continuous stream of problem-free solution upgrades."





## **Connecting a network of 2,500 student ambassadors with Google Workspace tools**

Even in normal times, communication across an archipelago of 17,500 islands, spanning 3,200 miles, can be a challenge. When Covid-19 hit, CICIL worried that the network of 2,500 volunteer student ambassadors it had built might become more disconnected, both from one another and the CICIL team in Jakarta.

Thanks to Google Workspace, student ambassadors were able to step up, rather than scale down, their mentoring activities across 50 Indonesian cities and 250 campuses during the pandemic. They conducted all teaching and coaching sessions, for groups of up to 30 students, via [Google Meet](https://workspace.google.com/products/meet/) (<https://workspace.google.com/products/meet/>) and found that engagement rose by 25% compared to live sessions, because of the ease of connecting online.

CICIL's core team was able to share educational files and programs safely and seamlessly with its external ambassador network on [Google Drive](https://workspace.google.com/products/drive/) (<https://workspace.google.com/products/drive/>), thanks

to Google Workspace's robust in-built security protocols.

Meanwhile, using Meet, Drive, and Google Calendar (<https://workspace.google.com/products/calendar/>), CICIL's staff moved 100% to remote working from home, with zero downtime in operations. "Google Workspace enabled us to operate at the same pace as usual, with no setbacks to business," says Jeremiah. "We're proud to say that this helped us get through the pandemic year with no salary cuts, budget cuts, or layoffs."

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## Looking beyond Indonesia with new AI-driven solutions

CICIL now plans to spread its vision of student empowerment beyond Indonesia to other countries in Southeast Asia where young people face similar challenges accessing student financing.

With limited resources for onboarding overseas staff, CICIL plans to enable its mission through adoption of more Google Cloud AI solutions. For example, it plans to deploy Cloud Vision (/vision) for optical character recognition (OCR) to read student ID cards and other nontraditional data sources, as well as analyzing student image assets, such as social media posts, for potential fraud detection. By automating such mundane tasks, CICIL's human team can then be freed up for creative endeavors, such as product and new market development.

"We have big ambitions for expanding across the region, but as a startup we can't simply go out and build teams in new markets," says Jeremiah. "With Google AI solutions such as Cloud Vision API,

there's no limit to our potential for growth even with a small team such as ours."

